

Feedback on Homework

We, the members of Group 2, have reviewed and evaluated the solutions submitted by Group 1 as follows:

Exercise 2.1

- ☒ Completed fully or nearly fully; results are correct or contain only minor inaccuracies (1 point)
- ☐ Substantially incomplete or fundamentally incorrect; only minimal relevant work has been provided (0 points)
- ☐ Not attempted; no relevant work has been submitted (0 points)

Exercise 2.2

- ☒ Completed fully or nearly fully; results are correct or contain only minor inaccuracies (1 point)
- ☐ Substantially incomplete or fundamentally incorrect; only minimal relevant work has been provided (0 points)
- ☐ Not attempted; no relevant work has been submitted (0 points)

Exercise 2.3

- ☒ Completed fully or nearly fully; results are correct or contain only minor inaccuracies (1 point)
- ☐ Substantially incomplete or fundamentally incorrect; only minimal relevant work has been provided (0 points)
- ☐ Not attempted; no relevant work has been submitted (0 points)

We have indicated within the completed assignments what was correct and where mistakes were made.

Further feedback (optional):

(c). Let $D_m = \text{diag}(\underbrace{1, \dots, 1}_{m-1}, -1, 1, \dots, 1)$

notice that $D_m A D_m = A$ ✓

For $i \neq j$, taking $m = i$, $(D_i A D_i)_{ij} = -a_{ij} = a_{ij} \Rightarrow a_{ij} = 0$

so A is a diagonal matrix, assume $A = \text{diag}(\lambda_1, \dots, \lambda_k)$

for $i \neq j$, consider a rotation matrix $R_{ij}(\theta)$

since $R_{ij}(\theta) A R_{ij}(\theta)^T = A$ ✓

$$\text{then } \begin{pmatrix} \lambda_i \cos^2 \theta + \lambda_j \sin^2 \theta & (\lambda_i - \lambda_j) \sin \theta \cos \theta \\ (\lambda_i - \lambda_j) \sin \theta \cos \theta & \lambda_i \sin^2 \theta + \lambda_j \cos^2 \theta \end{pmatrix} = \begin{pmatrix} \lambda_i & 0 \\ 0 & \lambda_j \end{pmatrix}$$

$\lambda_i = \lambda_j$ since i, j are arbitrary, $\lambda_1 = \dots = \lambda_k =: a$.

therefore $A = aI$.

(d). Since $A = aI$, $\sum_{i,j=1}^d a_{ij} u_{x_i} x_j = - \sum_{i=1}^d a u_{x_i} x_i = -a \Delta u$ ✓

(2). From (a), (b), b_i is constant and $Ub = b$, $\forall U \in O(d)$

let $U = I$, $b = -Ib = -b \Rightarrow b = 0$ ✓

then $\mathcal{L}u = -a \Delta u + cu$ □

Group 1

Wang Hao

(a) $(Lu)(x) = -\sum_{i,j=1}^d a_{ij}(x) u_{x_i x_j}(x) + \sum_{i=1}^d b_i(x) u_{x_i}(x) + c(x) u(x)$

Let $v(x) = u(x+y)$, then $v_{x_i}(x) = u_{x_i}(x+y)$, $v_{x_i x_j}(x) = u_{x_i x_j}(x+y)$

hence, $(Lu)(x+y) = -\sum_{i,j} a_{ij}(x+y) u_{x_i x_j}(x+y) + \sum_i b_i(x+y) u_{x_i}(x+y) + c(x+y) u(x+y)$

$$(Lv)(x) = -\sum_{i,j} a_{ij}(x) u_{x_i x_j}(x+y) + \sum_i b_i(x) u_{x_i}(x+y) + c(x) u(x+y)$$

$$(Lu)(x+y) = (Lv)(x)$$

therefore, for any x, y and any u .

$$-\sum_{i,j=1}^d (a_{ij}(x+y) - a_{ij}(x)) u_{x_i x_j}(x+y) + \sum_{i=1}^d (b_i(x+y) - b_i(x)) u_{x_i}(x+y)$$

$$+ (c(x+y) - c(x)) u(x+y) = 0$$

we obtain $a_{ij}(x+y) = a_{ij}(x)$, $b_i(x+y) = b_i(x)$, $c(x+y) = c(x)$

for all $x, y \in \mathbb{R}^d$, then all coefficients are constant.

(b) Let $A = (a_{ij})_{i,j=1}^d$, $v(x) = u(Ux)$, $U \in O(d)$

then $v_{x_i}(x) = \sum_{k=1}^d U_{ik} u_{x_k}(Ux)$, $v_{x_i x_j}(x) = \sum_{k,l=1}^d U_{ik} U_{jl} u_{x_k x_l}(Ux)$

thus $(Lv)(x) = -\sum_{i,j=1}^d a_{ij} v_{x_i x_j}(x) + \sum_{i=1}^d b_i v_{x_i}(x) + c u(Ux)$

$$= -\sum_{k,l=1}^d (UAU^T)_{kl} u_{x_k x_l}(Ux) + \sum_{k=1}^d (Ub)_k u_{x_k}(Ux) + c u(Ux)$$

$$(Lu)(Ux) = -\sum_{k,l=1}^d a_{kl} u_{x_k x_l}(Ux) + \sum_{k=1}^d b_k u_{x_k}(Ux) + c u(Ux)$$

$$(Lu)(Ux) = (L(u \circ U))(x) = (Lv)(x)$$

compare the 2nd-order term, we can get $UAU^T = A$ for all $U \in O(d)$

Group 1

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$$-\Delta u = 0 \quad (1)$$

$$u_t - \Delta u = 0 \quad (2)$$

$$-\Delta u = f \quad (3)$$

$$u_{tt} - \Delta u = 0 \quad (4)$$

(a) Determine coefficients $A = (a_{ij})_{i,j=1}^d$ of the second-order terms in the sense of (*).

① (Potential) $-\Delta u = 0$.

If $x = (x_1, \dots, x_d)$ the variables, then $-\Delta u = -\sum_{i=1}^d u_{x_i x_i}$.

So $A = -I_d = \begin{pmatrix} -1 & 0 & \dots & 0 \\ 0 & -1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & -1 \end{pmatrix}$. So it's elliptic.

② $u_t - \Delta u = 0$ (heat)

If $x = (t, x_1, \dots, x_{d-1})$, then $-\Delta u = -\sum_{i=1}^{d-1} u_{x_i x_i}$.

So $A = \text{diag}(0, -1, \dots, -1)$.

Since there is one eigenvalue and others have same sign, it's parabolic.

③ (Poisson) $-\Delta u = f$.

Since f doesn't affect the second-order term,

So $-\Delta u = -\sum_{i=1}^d u_{x_i x_i}$. So $A = -I_d$. Thus it's elliptic.

④ (Wave) $u_{tt} - \Delta u = 0$

Let $x = (t, x_1, \dots, x_{d-1})$, then second-order terms are

$$u_{tt} - \sum_{i=1}^{d-1} u_{x_i x_i}$$

So $A = \text{diag}(1, -1, \dots, -1)$.

Since eigenvalues have different signs, it's hyperbolic.

2.3 Consider 2-dim BVP

$$\begin{cases} \Delta u(x,y) = 0 & x \in \mathbb{R}, y > 0 \\ u(0,y) = 0 & y > 0 \\ u_y(x,0) = g(x) & x \in \mathbb{R} \end{cases}$$

Let $(g_n)_n$ be a seq. of boundary data given by
 $g_n(x) = n \exp(-\sqrt{n}) \sin(nx) \quad x \in \mathbb{R} \quad n \in \mathbb{N}$

(a) $u_n(x,y) = \exp(ny - \sqrt{n}) \sin(nx) \quad x \in \mathbb{R} \quad y \geq 0$

$$\frac{\partial u_n}{\partial x} = n \exp(ny - \sqrt{n}) \cos(nx)$$

$$\frac{\partial^2 u_n}{\partial x^2} = -n^2 \exp(ny - \sqrt{n}) \sin(nx)$$

$$\frac{\partial^2 u_n}{\partial y^2} = n^2 \exp(ny - \sqrt{n}) \sin(nx)$$

$$\Delta u_n = [-n^2 \exp(ny - \sqrt{n}) \sin(nx)] + [n^2 \exp(ny - \sqrt{n}) \sin(nx)] = 0$$

$$u_n(0,y) = \exp(ny - \sqrt{n}) \sin(0) = 0$$

$$\frac{\partial u_n}{\partial y}(x,0) = n \exp(n \cdot 0 - \sqrt{n}) \sin(nx) = g_n(x)$$